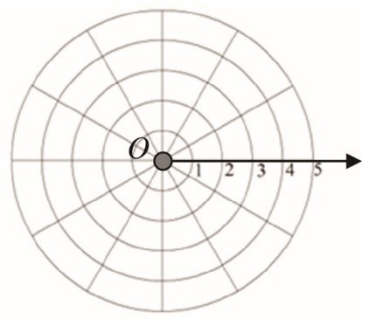
Plot points Using Polar Coordinates

A **polar coordinate system** consists of a fixed point O called the pole (or origin), and a ray called the polar axis, with endpoint at the pole.



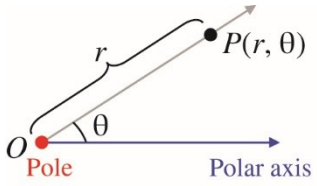
Each point P in the plane is defined by an ordered pair of the form (r, θ) where r is the directed distance from the pole to P.

If $r > 0$, point P is located r units from the pole in the direction of θ.

If $r < 0$, point P is located $|r|$ units from the pole in the direction of $\theta + \pi$ (the direction *opposite* θ).

If $r = 0$, point P is located at the pole.

θ is a directed angle from the polar axis to line $\overset{\frown}{OP}$.



$\theta > 0$ is measured counterclockwise from the polar axis.

$\theta < 0$ is measured clockwise from the polar axis.

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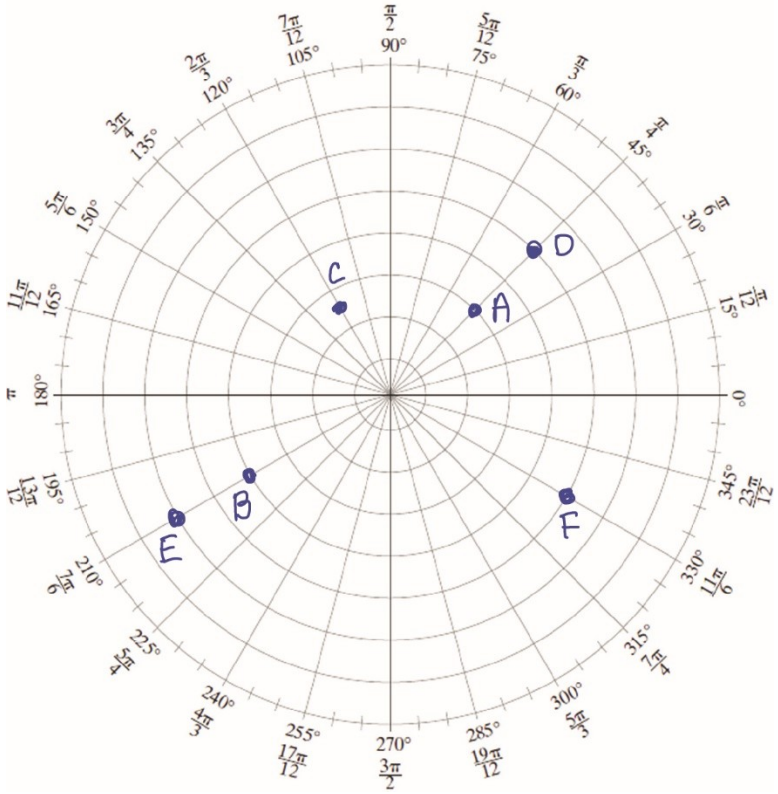
1. Plot the points.

$\left(3, \frac{\pi}{4}\right)$ $\left(-4, \frac{\pi}{6}\right)$

A B

C $\left(2.5, -\frac{4\pi}{3}\right)$ D $\left(5, \frac{9\pi}{4}\right)$

E $\left(6, \frac{7\pi}{6}\right)$ F $\left(-5, \frac{5\pi}{6}\right)$



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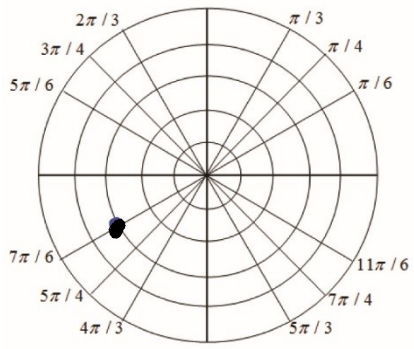
use this when $r > 0$

The ordered pairs (r, θ) and $(r, \theta + 2n\pi)$ for any integer n represent the same point in polar coordinates.

2. Given the point $\left(3, \frac{7\pi}{6}\right)$ Find two other polar coordinate representations with $r > 0$.

$\frac{7\pi}{6} + 2\pi = \frac{19\pi}{6}$ $\left(3, \frac{19\pi}{6}\right)$

$\frac{7\pi}{6} - 2\pi = -\frac{5\pi}{6}$ $\left(3, \frac{-5\pi}{6}\right)$



use this when $r < 0$

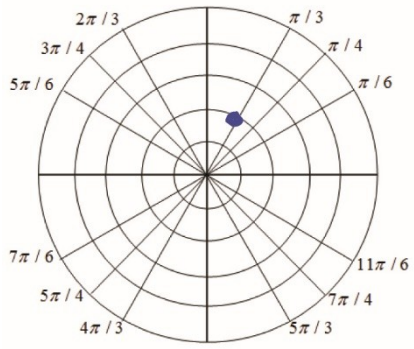
The ordered pairs (r, θ) and $(-r, \theta + (2n+1)\pi)$ for any integer n represent the same point.

$\theta + (2n+1)\pi$ represents θ plus or minus an odd multiple of π. This is an angle with terminal side in the opposite direction of θ.

3. Given the point $\left(2, \frac{\pi}{3}\right)$ Find two other polar coordinate representations with $r < 0$.

$\frac{\pi}{3} + \pi = \frac{4\pi}{3}$ $\left(-2, \frac{4\pi}{3}\right)$

$\frac{\pi}{3} - \pi = -\frac{2\pi}{3}$ $\left(-2, \frac{-2\pi}{3}\right)$



Convert Ordered Pairs Between Polar and Rectangular Coordinates

Convert polar coordinates (r, θ) to rectangular coordinates (x, y): $x = r\cos\theta$ and $y = r\sin\theta$

Convert rectangular coordinates (x, y) to polar coordinates (r, θ): $r^2 = x^2 + y^2$ and $\tan\theta = \frac{y}{x}$ for $x \neq 0$

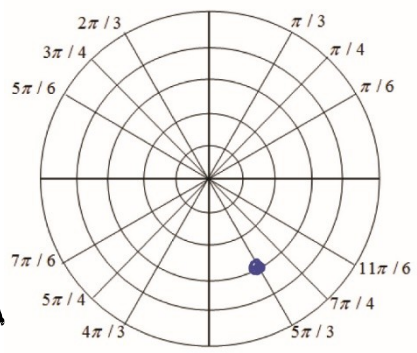
Convert the ordered pair in polar coordinates to rectangular coordinates.

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4. $\left(3, \frac{5\pi}{3}\right)$ $x = 3\cos\frac{5\pi}{3} = \frac{3}{2}$ $x = r\cos\theta$

$y = 3\sin\frac{5\pi}{3} = \frac{-3\sqrt{3}}{2}$ QIV $y = r\sin\theta$

rectangular coords → $\left(\frac{3}{2}, \frac{-3\sqrt{3}}{2}\right)$



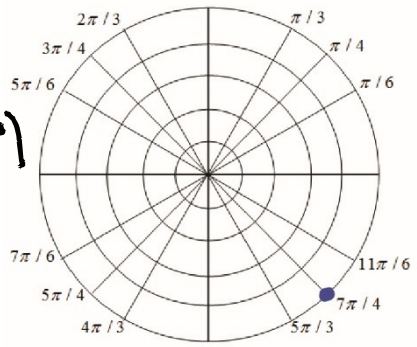
5. $\left(5, -\frac{\pi}{4}\right)$ $x = 5\cos\left(\frac{-\pi}{4}\right) = \frac{5\sqrt{2}}{2}$

$y = 5\sin\left(\frac{-\pi}{4}\right) = \frac{-5\sqrt{2}}{2}$ QIV

$\left(\frac{5\sqrt{2}}{2}, \frac{-5\sqrt{2}}{2}\right)$

Polar coordinates will always be in terms of pi or degrees

$(3, \frac{5\pi}{3})$ or $(3, 330°)$



Convert the ordered pair in rectangular coordinates to polar coordinates with $r > 0$ and $0 < \theta < 2\pi$.

6. $\left(-3\sqrt{2}, 3\sqrt{2}\right)$

$r^2 = x^2 + y^2$

$r^2 = \left(-3\sqrt{2}\right)^2 + \left(3\sqrt{2}\right)^2$

$r^2 = 36,\ r = \pm 6$

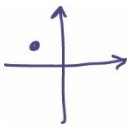
sign of r depends on this expression

$\tan\theta = \frac{y}{x}$

$\tan\theta = \frac{3\sqrt{2}}{-3\sqrt{2}} = -1$

$\theta = \frac{3\pi}{4}$

$\left(6, \frac{3\pi}{4}\right)$



7. $(2, -4)$

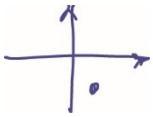
$r^2 = 2^2 + (-4)^2$

$r^2 = 20, r = \pm 2\sqrt{5}$

$\tan\theta = \frac{-4}{2} = -2,\ \theta = \tan^{-1}(-2)$ in QIV

or $2\pi + \tan^{-1}(-2)$

$\left(2\sqrt{5}, 2\pi + \tan^{-1}(-2)\right)$



Convert Between Equations in Polar and Rectangular Coordinates

Convert equations from rectangular coordinates to polar coordinates by substituting $x = r\cos\theta$ and $y = r\sin\theta$.

To convert an equation from polar coordinates to rectangular coordinates, we can use one or more of the following strategies.

1. Apply one or more of the relationships $x = r\cos\theta$, $y = r\sin\theta$, $x^2 + y^2 = r^2$, or $\tan\theta = \frac{y}{x}$ for $(x \neq 0)$.
2. Multiply both sides by r.
3. Square both sides of the equation.

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Write an equivalent equation using polar coordinates.

$x = 12$

$r\cos\theta = 12$

$r = \dfrac{12}{\cos\theta}$

8. $r = 12\sec\theta$

$y = -6$

$r\sin\theta = -6$

$r = \dfrac{-6}{\sin\theta}$

9. $r = -6\csc\theta$

$x^2 + y^2 = 121$ $\quad x^2 + y^2 = r^2$

$r^2 = 121$

$r = \pm\sqrt{121}$

$r = 11$

$2x - 5y = 4$

$2r\cos\theta - 5r\sin\theta = 4$

$r(2\cos\theta - 5\sin\theta) = 4$

$r = \dfrac{4}{2\cos\theta - 5\sin\theta}$

$x = 4y^2$

$r\cos\theta = 4(r\sin\theta)^2$

$\dfrac{r\cos\theta}{4r\sin^2\theta} = \dfrac{4r^2\sin^2\theta}{4r\sin^2\theta}$

$\dfrac{\cos\theta}{4\sin^2\theta} = r$

$\dfrac{1}{4}\cdot\dfrac{\cos\theta}{\sin\theta}\cdot\dfrac{1}{\sin\theta} = r$

$\dfrac{1}{4}\cdot\cot\theta\cdot\csc\theta = r$

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10. $2x-5y=4$

$2r\cos\theta-5r\sin\theta=4$

$r(2\cos\theta-5\sin\theta)=4$

$r=\dfrac{4}{2\cos\theta-5\sin\theta}$

11. $x^2+y^2=121$

$r^2=121$

$r=11$

12. $x=4y^2$

$\dfrac{r}{r}\cos\theta=\dfrac{4r^2}{r}\sin^2\theta$

$\cos\theta=4r\sin^2\theta$

$\dfrac{1}{4}\dfrac{\cos\theta}{\sin^2\theta}=r\rightarrow\dfrac{1}{4}\dfrac{\cos\theta}{\sin\theta}\cdot\dfrac{1}{\sin\theta}=r$

$(r\neq 0)$ $r=\dfrac{1}{4}\cot\theta\csc\theta$

13. $(x-5)^2+y^2=25$

$(r\cos\theta-5)^2+r^2\sin^2\theta=25$

$r^2\cos^2\theta-10r\cos\theta+25+r^2\sin^2\theta=25$

$r^2\left(\cos^2\theta+\sin^2\theta\right)-10r\cos\theta=0$ $=1$

$r^2-10r\cos\theta=0$

$r(r-10\cos\theta)=0$

$r=0$ $\quad r=10\cos\theta$ $\quad (r=0$ is included in $r=10\cos\theta)$

Convert the polar equation to rectangular form and identify the type of curve represented.

14. $r=15$

$r^2=225$

$x^2+y^2=225$ circle

15. $\theta=\dfrac{2\pi}{3}$

$\tan\theta=\tan\dfrac{2\pi}{3}$

$\dfrac{y}{x}=-\sqrt{3}$

$y=-\sqrt{3}x$ line

$y=mx+b$

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16. $r = -8\sec\theta$

$$r = -\frac{8}{\cos\theta}$$

$$r\cos\theta = -8$$

$x = -8$ vertical line

17. $r = 10\sin\theta$

$$r^2 = 10r\sin\theta$$

$$x^2 + y^2 = 10y$$

$$x^2 + y^2 - 10y + 25 = 0 + 25$$

$x^2 + (y-5)^2 = 25$ circle



review this one a shit ton

$(a-b)^2 = a^2 - 2ab + b^2$

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